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Why Skordis and Złóśnik Were Right: The MOND Acceleration Scale as a de Sitter–Unruh Manifestation of the Cosmological Constant

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Version 2026-06-20 · Zenodo [10.5281/zenodo.20772880](https://zenodo.org/record/20772880). Companion to the published framework (Zenodo [10.5281/zenodo.20721540](https://zenodo.org/record/20721540)) and the evolving- $a_0(z)$ note (Zenodo [10.5281/zenodo.20737162](https://zenodo.org/record/20737162)). *Every quantitative claim below is reproduced by a committed Python script; the paths are listed in §9 and all 60 scripts exit 0.*

Abstract

The Aether-Scalar-Tensor theory (AeST) of Skordis and Złóśnik (2021) is the relativistic completion of MOND that fits the full *Planck* cosmic-microwave-background power spectrum, including the third acoustic peak, while reproducing the galaxy-scale radial acceleration relation. It has, however, been criticised on three grounds: (i) its scalar potential — a “ghost-condensate” kinetic function $K(Q)=\mu^2(Q-1)^2$ — is *postulated*; (ii) its preferred (aether) frame is *postulated*; and (iii) its MOND acceleration scale a_0 is *fitted*, not derived. We argue that on every count Skordis and Złóśnik were *right*, and that the framework $a_0 = c^2 \sqrt{(\Lambda/32\pi)} = (c/2) \sqrt{(G\rho DE)} = cH\Lambda/Z$ (with $Z = \sqrt{(32\pi/3)} = 5.7888$) supplies precisely the physical motivation those three postulates were missing — without altering a single one of AeST’s successful predictions. The acceleration scale is *tied to the cosmological constant* through a de Sitter–Unruh effective temperature, fixing $a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$ to within the radial-acceleration-relation band; the preferred frame is *named* by the de Sitter vacuum (the cosmic rest frame where the Gibbons–Hawking floor is isotropic); and the $K(Q)$ ghost-condensate is shown to be internally healthy — it clears the boost-free infrared positivity/causality bound of Grall–Melville and Serra–Trombetta (the latter written at Skordis’s own institute). We are scrupulously two-sided: the framework **derives no number** (a_0 ’s coefficient, the kinetic-term origin, and the dark-sector amplitude all remain free or postulated), its cluster-core deficit is **shared**

identically with AeST, and the positivity result is a **saturation-pass that transmits no new constraint**. The net is a vindication, not a replacement: AeST is sound, well-founded, and — read as modified *inertia* with a_0 set by Λ — even better-motivated than its authors claimed. We close with the live falsifiable tests (Gaia DR4 wide binaries, the $\hat{s}^{\text{TX}}$ Lorentz-violation dipole, the evolving- a_0 Tully–Fisher offset) whose first verdict arrives in ~December 2026.

1. Introduction

Modified Newtonian Dynamics (MOND; Milgrom 1983) reproduces galaxy rotation curves and the baryonic Tully–Fisher relation from a single acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$, with no galaxy-scale dark matter. Its long-standing obstacle was a relativistic completion compatible with cosmology and gravitational lensing. The Aether-Scalar-Tensor theory (AeST; Skordis & Złośnik 2021, *Phys. Rev. Lett.* **127**, 161302, arXiv:2007.00082) cleared that obstacle: a unit-timelike aether A_μ plus a shift-symmetric scalar with a kinetic function $K(Q)$ and a MOND function $J(Y)$ reproduce both the MOND limit *and* the full *Planck* CMB power spectrum, third peak included, by carrying a cold, dust-like energy density that mimics cold dark matter on linear scales.

AeST is nonetheless commonly described as having three “free” ingredients: the scalar kinetic function $K(Q)=\mu^2(Q-1)^2$, the preferred aether frame, and the value of a_0 . This paper’s thesis is that these are not weaknesses of AeST but *open slots*, and that a single physical input — **the cosmological constant Λ acting through a de Sitter–Unruh effective temperature** — fills all three, fixing the framework

$$a_0 = c^2 \sqrt{\frac{\Lambda}{32\pi}} = \frac{c}{2} \sqrt{G\rho_{\text{DE}}} = \frac{cH_\Lambda}{Z}, \quad Z = \sqrt{\frac{32\pi}{3}} = 5.7888,$$

to $a_0 = 9.3624 \times 10^{-11} \text{ m s}^{-2}$ for $\Lambda = 1.0909 \times 10^{-52} \text{ m}^{-2}$ ($\Omega_\Lambda = 0.685$, $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$). This value sits inside the radial-acceleration-relation (RAR) optimum band ($\approx 7.5 \times 10^{-11}$ – $1.8 \times 10^{-10} \text{ m s}^{-2}$, $\leq 2\%$ scatter penalty across interpolation/M-L conventions); it is *not* a new fit but a re-reading of AeST’s existing a_0 as a manifestation of Λ . Three independent reproductions of the value are in [reviews/ghost_condensate/gc_scale_check.py](#).

We honour a strict both-ways discipline throughout: a claimed deficit is verified as rigorously as a claimed success, on the framework’s *own* footing ($a_0 = 9.36 \times 10^{-11}$, the de Sitter–Unruh interpolation $g_{\text{obs}} = \sqrt{(g_{\text{bar}}^2 + g_{\text{bar}} \cdot a_0)}$, stellar mass-to-light $\Upsilon \approx 0.7$), never the regular-MOND defaults. The quarantine is absolute: **a_0 , Z , the coefficient $\kappa = \frac{1}{2}$, and the dark-sector amplitude I_0 are never asserted to be *derived*.**

2. The acceleration scale from the cosmological constant

A test body of proper acceleration $\mathbf{a}$ immersed in the de Sitter background of a universe with cosmological constant Λ experiences a Deser–Levin effective temperature

$$T_{\text{eff}} = \frac{\hbar}{2\pi c k_B} \sqrt{a^2 + (cH_\Lambda)^2}, \quad cH_\Lambda = c^2 \sqrt{\Lambda/3},$$

interpolating between the Unruh temperature ($a \gg cH_\Lambda$) and the Gibbons–Hawking temperature ($a \ll cH_\Lambda$). Reading inertia as the response to this vacuum (modified-inertia MOND), the transition between Newtonian and deep-MOND dynamics occurs where the two terms are comparable,

i.e. at an acceleration of order $cH_0\Lambda$. The geometric coefficient that converts $cH_0\Lambda$ into a_0 is $Z = cH_0\Lambda/a_0 = \sqrt{32\pi/3}$. Its content is transparent once a_0 is written in density form: from $\rho_{\text{DE}} = \Lambda c^2/(8\pi G)$ one has $\sqrt{G\rho_{\text{DE}}} = c\sqrt{\Lambda/8\pi}$, so

$$a_0 = \frac{c}{2}\sqrt{G\rho_{\text{DE}}} = c^2\sqrt{\frac{\Lambda}{4\cdot 8\pi}} = c^2\sqrt{\frac{\Lambda}{32\pi}}, \quad Z^2 = \frac{32\pi}{3} = \frac{4\cdot 8\pi}{3},$$

so that **8π is Einstein’s field-equation normalisation, 3 is the Friedmann factor ($H_0\Lambda^2 = \Lambda c^2/3$), and the remaining 4 is the inverse-square of the coefficient $\kappa = \frac{1}{2}$ in $a_0 = \kappa c\sqrt{G\rho_{\text{DE}}}$ — pure spacetime geometry up to that one free κ . The ****coefficient $\kappa = \frac{1}{2}$ in $a_0 = \kappa c\sqrt{G\rho_{\text{DE}}}$ is a free input**** — it is not forced by ghost-freedom, unitarity, or holography (each is κ -invariant), and we re-confirm below (§5, Door S) that no independent mechanism transmits it. This is the framework’s defining equation and its single quarantined number; what follows treats it as *given* and asks whether AeST is the right home for it.**

3. The AeST embedding: the dark sector is the framework’s own field

The framework’s modified-inertia reading and AeST’s modified-gravity reading share one scalar field φ . Its spatial gradient mode (the J(Y) sector) supplies the a_0 -scale MOND force; its temporal mode (the K(Q) sector) supplies a cold, a^{-3} “dust” energy that mimics dark matter — *not a particle, a mode of the gravitational field itself*. The kinetic function

$$K(Q) = \mu^2(Q-1)^2$$

is, structurally, a **ghost condensate** in the sense of Arkani-Hamed, Cheng, Luty & Mukohyama (2004, hep-th/0312099): a $P(X)$ with a non-trivial minimum at $X_0 > 0$. This is the *authors’ own* identification — Verwayen, Skordis & Złótnik (2024) write Eq. (7) as exactly $\mu^2(Q-1)^2$ and cite ACLM; Blanchet & Skordis (2024, *JCAP* **11**, 040, arXiv:2404.06584) adopt the identical $K(Q)$ in their “Relativistic Khronon” action. The structural mapping is sympy-verified in [reviews/ghost_condensate/map_aest_to_ghost.py](#) and [expand_PX_around_condensate.py](#): $K(1)=0$, $K'(1)=2\mu^2>0$ (a true minimum, no ghost), and the shift-charge first integral $a^3K(Q)=I_0$ gives the a^{-3} dust.

Two of AeST’s three “postulates” acquire physical content from §2:

- **The preferred frame is named, not postulated.** The framework’s lensing no-go (co-variant, Cassini-safe MOND lensing is forbidden by diffeomorphism invariance with $c_T=c$ and ghost-freedom) *forces* a preferred frame; a preferred unit-timelike frame *is* an aether A_μ . The de Sitter–Unruh vacuum then selects *which* frame: the cosmic rest frame, where T_{eff} has its isotropic Gibbons–Hawking floor. AeST postulates the aether and justifies it a posteriori by the CMB fit; the framework supplies the missing front-end motivation.
- **The dark-sector “condensate” evades the symmetry obstruction that killed vacuum induction.** A de Sitter *vacuum* is $SO(4,1)$ -invariant and induces no preferred-frame kinetic term; the condensate breaks the symmetry through a *background solution*, not the vacuum (the ferromagnet/magnon mechanism), and the same de Sitter background cures the ghost-condensate Jeans instability by Hubble friction ($H_0/\Gamma \sim 10^{25}\text{--}10^{31}$; [reviews/door_runs/C_lyapunov_stability.py](#)).

That $a_0 \propto \sqrt{\Lambda}$ is not an artefact of one mechanism: Blanchet & Seraille (2025, *JCAP* **12**, 036, arXiv:2502.14686) obtain the *same* scaling $a_0 \sim c^2 \sqrt{\Lambda}$ from an entirely independent SU(2) Yang–Mills graviphoton theory. The *form* is multiply realised; the coefficient is not (§5, Door S).

4. Consistency: AeST’s kinetic term is healthy

The sharpest theoretical worry about a ghost-condensate dark sector is whether its wrong-sign-then-stabilised kinetic structure violates positivity/causality. The applicable bounds are the **boost-free** ones — Grall & Melville (2021, *Phys. Rev. D* **105**, L121301, arXiv:2102.05683) and Serra & Trombetta (2024, arXiv:2412.19745) — which rest on unitarity, causality and locality, *not* Lorentz invariance, and therefore bind a Lorentz-violating condensate. (The stronger Creminelli–Janssen–Senatore bounds require a conformal UV completion the condensate lacks and are *inapplicable* — they must not be cited as a kill.) Notably, Trombetta is at CEICO Prague, Skordis’s own institute: the theorem that adjudicates the framework’s positivity was written by the home cluster. We are careful about attribution: Skordis and Złośnik themselves established AeST’s ghost-freedom and its linear stability window $\{0 < K_B < 2, \mu^2 > 0, \lambda_s > 0\}$ in 2020–21, and we credit that result in full; the boost-free positivity bounds applied below *postdate* AeST (2021, 2024), so what follows is an independent later confirmation of their already-stable action, not a substitute for it.

We report two results, both ways.

(a) **The sign of the dark-matter mode is forced.** Energy-positivity of the dust ($\rho_dust \propto +4\mu^2 \cdot \delta Q > 0$) forces the shift-charge $\text{sign}(I_0) > 0$, hence $Q > 1$, which is exactly the gradient-stable branch $c_s^2 = \delta Q / (3\delta Q + 2) > 0$. Energy-positivity and sound-speed-positivity *coincide* on the dust branch: **the dark mode is gradient-stable by the very requirement that makes it dark.** This is sympy-exact in [reviews/door_runs/doorA_positivity_gate.py](#) and [condensate_postulate_and_eos.py](#). It is a *new constraint on the theory’s initial condition*, and it survives independently of the dispersion details.

(b) **The positivity bound is cleared — by saturation, adding no new constraint (stated plainly).** Reading the *real* Blanchet–Skordis quadratic action (not a generic ghost-condensate placeholder), the propagating scalar has dispersion $\omega = 0$ at quadratic order — there is **no propagating k^4 tail** (Blanchet–Skordis Sec. 6.2: “there are no higher-derivative interaction terms ... quadratic in the fields”); MOND emerges from the *nonlinear* term. The faithful gapped/gapless pair is same-sector and shares one sound speed, so the Serra–Trombetta ratio v^2_gapped/c_s^2 is *identically* across the entire stability window $\{0 < K_B < 2, K_2 > 0, \lambda_s > 0\}$. **Positivity/causality is cleared everywhere — but by equality, sitting on the bound, and it carves out no new parameter sub-window.** We explicitly *retract* an earlier (placeholder-driven) suggestion of a K_B exclusion; there is none ([reviews/door_runs/doorA_real_coefficients.py](#)). The honest reading: a genuine consistency *pass*, not a discovery, and a reminder that the ghost-condensate k^4 picture is a useful EFT analogy, not the host’s literal quadratic structure.

(c) **Lensing reconciles with the host.** The framework’s gravitational slip $\gamma = 2 \sqrt{(1+a_0/g_N)} - 1$ is *not* a field-level discriminator against AeST: it is the effective-dark-matter “inferred slip” (the ratio of the real phantom-mass lensing potential to twice the baryon-only Newtonian potential), which the Blanchet–Skordis khronon produces identically (their $\varphi=\psi$ at Eq. 3.10). The null-geodesic deflection $\alpha(b)$ is identical to the host to a ratio of 1.0000 at every impact parameter ([reviews/door_runs/door_L_lensing_headtohead.py](#)). The slip survives only as a MOND-family signature versus Λ CDM (which has $\gamma=1$), not as a framework-versus-AeST discriminator.

5. The honest ledger: established, shared, open

#	Sector	Verdict (framework footing)	Script
1	Galaxies / RAR / BTFR	PASS (AeST MOND limit; $a_0 = c^2 \sqrt{(\Lambda/32\pi)}$ is the scale)	published framework
2	CMB + linear P(k)	PASS (AeST fits <i>Planck</i> ; dust $\approx$ CDM on linear scales)	Skordis– Złošnik 2021
3	Lensing	PASS (GR-of-the-phantom-mass; agrees with the host, §4c)	door_L_lensing_headtohead
4	Cassini / Solar System	PASS ($c_T=c$; MOND screened where $g_N \gg a_0$)	published framework
5	Ghost-freedom / 6-dof stability	PASS (windowed $\{K_B, K_2, \lambda_s\}$)	C_lyapunov_stability.py
6	IR positivity / causality	PASS by saturation (no new exclusion, §4b)	doorA_real_coefficients.p
7	a_0 value (κ)	OPEN — free (Door S: Seraille’s $\sqrt{\Lambda}$ form matches, coefficient 32π not transmitted)	door_S_seraille_coefficie
8	Kinetic-term <i>origin</i>	OPEN — postulated (Door D, Oda both NOT-FORCED, §6)	D_mersini_forcing.py, oda_door/
9	Dark-sector <i>amount</i> (Ω_{dm})	OPEN — free (a shift-charge integration constant)	door_E_amount_pin.py
10	Cluster cores	SHARED MOND deficit (identical to AeST, §6)	clash_cluster/

The framework **closes no gap**: a_0 ’s value, the kinetic-term origin, and the amount are all free or postulated — *exactly as they are in AeST itself*. The vindication is that all of AeST’s successes are reproduced and its kinetic term is shown healthy, while none of the framework’s distinctive content manufactures a derivation it does not have.

6. What the framework does *not* do (both ways)

Cluster cores — a shared, not a framework-specific, failure. On the precise CLASH weak-lensing cluster-core residual (Famaey, Pizzuti & Saltas 2024, arXiv:2410.02612) the framework’s modified-inertia phantom under-supplies the cored residual by a factor ≈ 4.4 (core-averaged 100–450 kpc), recovered identically by the eRASS1 X-ray core (ratio 1.03 between the two probes). Crucially, the framework’s modified-*inertia* residual equals AeST’s modified-*gravity* residual to machine precision (deep-MOND scale invariance pins $M \cdot G \cdot a_0 = \eta \cdot \sigma^4$ with $\eta \sim 1$, shared); AeST itself undershoots cluster cores (Durakovic & Skordis 2023, arXiv:2312.00889). The same deficit appears in the new JWST strong-lensing cluster XLSSC 122 at $z=1.98$ (Finner et al. 2025), where the framework undershoots the concentrated $M(<100 \text{ kpc})=6.5 \times 10^{13} \text{ M}$ core by $\sim 6\text{--}20\times$. This is the 40-year MOND cluster-core problem, *shared across the relativistic-MOND family* and not closeable without an additional component; it is not a referee-proof kill given the post-XRISM equilibrium ambiguity, but we report it as an open deficit, not a manufactured win. Scripts: [reviews/clash_cluster/framework_mi_residual.py](#), [aest_mg_vs_mi_residual.py](#).

The kinetic-term origin is not derived. Three independent attempts to *derive* the form of $K(Q)$ from Λ all return NOT-FORCED: the de Sitter–Unruh vacuum-induction route (killed by a sympy-proven theorem — the defining representation of $\mathfrak{so}(4,1)$ is irreducible, so the de Sitter vacuum admits *no* invariant timelike vector and can induce no preferred frame; a static-patch observer or the condensate background evades it by breaking the symmetry with a *state*, recovering the frame as an $\text{SO}(3)$ -singlet Goldstone but as a postulated VEV, never a derived stiffness — [reviews/gap2_rep_door/so41_no_invariant_timelike_vector.py](#)); the Mersini-Houghton phantom $\Rightarrow$ time-crystal forcing theorem (vacuous here — the framework is never phantom, $w \in (-1, \frac{1}{3})$; [reviews/door_runs/D_mersini_forcing.py](#)); and the most direct published “ghost condensate from Λ ” mechanism, Oda (2025, arXiv:2509.23648), which turns out to be a *false cognate* — its “ghost” is a fermionic Faddeev–Popov gauge ghost generating only Einstein–Hilbert gravity, structurally excluding the bosonic scalar the framework’s dark sector is, and leaving a_0 untouched since $\partial a_0 / \partial G = 0$ ([reviews/oda_door/oda_gap1_mapping.py](#)). The kinetic term remains *postulated* — as it is in AeST.

The amount is free. No de Sitter number pins $\Omega_{dm}/\Omega_\Lambda = 0.388$; the dust amplitude I_0 is a free shift-charge integration constant ($d\rho_{\text{dust}}/d\Lambda = 0$ at the level of the FRW equations; [reviews/door_runs/door_E_amount_pin.py](#)), in agreement with the independent “Ghost Dark Matter” construction of Lim, Sawicki & Vikman (2010, arXiv:1001.4634).

7. Falsifiable predictions and live tests

The framework inherits AeST’s $z=0$ record and adds one distinctive content — a *declining* acceleration scale tied to the dark-energy density, $a_0(z)/a_0(0) = \sqrt{(\rho_{\text{DE}}(z)/\rho_{\text{DE}}(0))}$, which under the DESI DR2 CPL parameters ($w_0=-0.752$, $w_a=-0.86$) gives $a_0(z=2) \approx 0.86 a_0(0)$, $a_0(z=3) \approx 0.74 a_0(0)$. The decisive tests, with the first verdict imminent:

Test	What it decides	Earliest verdict
Gaia DR4 wide binaries	the MOND premise at $z=0$	~Dec 2026
$\hat{s}^{\text{TX}}$ SME dipole	the preferred-frame (Lorentz-violation) prediction ($\sim 9.6\times$ margin)	$\sim 2028\text{--}2032$
evolving- a_0 Tully–Fisher offset	the <i>distinctive</i> declining- a_0 (discs $\sim 7\%$ below the $z=0$ BTFR)	DESI DR3 + ELT/JWST, early-mid 2030s
cluster non-adiabatic σ -spread (plunging UDG/dSph, outer MOND zone)	modified <i>inertia</i> specifically: internal σ rises with infall phase $\sim 6\text{--}13\%$ (MG exactly 0 , a machine-verified theorem; CDM tidal fakes $\sim 2\text{--}8\%$ but radially anti-correlated)	ELT resolved- σ , $\sim 2032+$

If $w(z) \rightarrow -1$ (no evolving dark energy), the distinctive $a_0(z)$ content vanishes and the framework degenerates to constant- a_0 AeST. The cluster-core deficit is *not* a discriminator — it is shared with AeST and “explained” by ΛCDM with particle dark matter.

8. Conclusion: why Skordis and Złośnik were right

AeST was criticised for postulating its kinetic function, its preferred frame, and its acceleration scale. We have argued that all three postulates are *correct slots* filled by one physical input — the cosmological constant acting through a de Sitter–Unruh effective temperature. The acceleration scale is fixed to $a_0 = c^2 \sqrt{(\Lambda/32\pi)} = 9.36 \times 10^{-11} \text{ m s}^{-2}$ (within the RAR band); the preferred frame is the cosmic rest frame the de Sitter vacuum names; and the K(Q) ghost-condensate is internally healthy — energy-positivity forces the dark mode onto its gradient-stable branch, and the boost-free positivity/causality bound (from Skordis’s own institute) is cleared.

We have been equally clear about the limits. The framework derives no number it did not already postulate; its positivity pass is a saturation, not a margin; its cluster-core deficit is real and shared identically with AeST; and three independent attempts to derive the kinetic term from Λ all fail. The honest standing is a *well-founded effective theory at a real frontier* — not a completed theory of everything, and not numerology.

That standing is, in our reading, a vindication of Skordis and Złośnik. AeST is the right relativistic MOND; tying its acceleration scale to the cosmological constant and reading it as modified inertia makes it better-motivated, fully consistent with the boost-free positivity programme, and falsifiable on a near-term timeline — without disturbing a single one of its successful predictions. The first datum that could overturn it all arrives, fittingly, within the year.

9. Computational reproducibility

All quantitative claims are reproduced by committed Python scripts under `opus_48_extended_research/reviews`, all 60 scripts exit 0 (verified 2026-06-19). The load-bearing files:

- **Acceleration scale / value** — `ghost_condensate/gc_scale_check.py`, `ghost_condensate/dsunruh_dri`
- **Ghost-condensate mapping** — `ghost_condensate/map_aest_to_ghost.py`, `expand_PX_around_conden`, `condensate_postulate_and_eos.py`, `seesaw_two_scales.py`, `amount_and_pathologies_calc.py`, `adversarial_gate_and_eos.py`
- **Consequences (dispersion, S8, EFT window)** — `gc_consequences/pin_I0_thermal.py`, `pk_k4_signature.py`, `w_gradient_cmb_calc.py`, `w_gradient_cmb_scherrer.py`, `eft_validity_window_c`
- **Positivity (Door A) + pin** — `door_runs/doorA_positivity_gate.py`, `doorA_real_coefficients.py`, `doorA_real_coefficients_SKEPTIC.py`, `doorA_REAL_coefficients_RATIO_SCAN.py`, `doorA_REAL_coefficients_VERIFY.py`
- **Origin (Door D, Oda, GAP-2 frame)** — `door_runs/D_mersini_forcing.py`, `oda_door/oda_gap1_mapping.py`, `oda_door/oda_steelman_bothways.py`, `gap2_rep_door/so41_no_invar` (the $SO(4,1)$ no-invariant-timelike-vector theorem)
- **Lensing (Door L)** — `door_runs/door_L_lensing_headtohead.py`, `door_L_lensing_headtohead_SKEPTI`
- **K(Q) family, stability, amount, coefficient, window** — `door_runs/doorB_quadratic_vs_dbi.py`, `C_lyapunov_stability.py`, `door_E_amount_pin.py`, `door_S_seraille_coefficient.py`, `doorF_aclm_window.py`
- **Clusters** — `clash_cluster/clash_target_profile.py`, `framework_mi_residual.py`, `aest_mg_vs_mi_residual.py`, `robustness_bothways.py`
- **Distinctive σ -spread prediction** — `sigma_predict/mi_amplitude_band.py`, `mg_zero_theorem.py`, `cdm_tidal_confound.py`, `verify_MG_zero_and_CDM_separation.py`, `observational_test_design.py`

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Both-ways discipline and quarantine ($a_0/Z/\kappa/I_0$ never asserted derived) maintained throughout. This is a synthesis/position paper, not a discovery paper: it claims a re-reading of AeST’s acceleration scale and a consistency demonstration, not a new derivation.